

THEME 3: DIFFERENTIATION

Unit 3.1 Derivatives of polynomials and exponential functions

Source: Stewart, Section 3.1 pp 174 - 184.

Learning objectives

On completion of this unit you should

1. be able to write down and use the rule to find the derivative of a constant function, power functions, the natural exponential function, constant multiple of a function, sum of functions, difference of functions;
2. be able to use the rules and formulas in problems.

Remark

You do not have to prove any of the rules in this unit.

Assigned Problems

Stewart, Exercises 3.1 p 181: All odd numbers from 3 to 35, 50, 53, 59, 60, 61, 62, 65, 67, 71, 78, 85.

Unit 3.2 The product rule and the quotient rule

Source: Stewart, Section 3.2 pp 185 - 191.

Learning objectives

On completion of this unit you should

1. be able to write down and use the rule to find the derivative of the product of functions;
2. be able to write down and use the rule to find the derivative of the quotient of functions;
3. be able to use the rules in problems.

Assigned Problems

Stewart, Exercises 3.2 p 189: All odd numbers from 3 to 25, 33, 39(a), 43, 45, 53, 55, 66.

Unit 3.3 Derivatives of trigonometric functions

Source: Stewart, Section 3.3 pp 191 - 199.

Learning objectives

On completion of this unit you should

1. be able to write down the derivatives of the six trigonometric functions;
2. be able to use these derivatives in problems.

Remarks

1. You should be able to prove that $\frac{d}{dx} \sin x = \cos x$ and that $\frac{d}{dx} \cos x = -\sin x$ by using the limit definition.

You may assume that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, and do not have to know the proof on p 195.

You must be able to prove that $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$ (p 196).

2. You should be able to prove the formulas for the derivatives of the other four trigonometric functions by using the derivatives of sin and cos and the quotient rule.

Assigned Problems

Exercises 3.3 p 197: All odd numbers from 9 to 27, 32(a), 34(a), 39, 41, 43.

Unit 3.4 The chain rule

Source: Stewart, Section 3.4 pp 199 - 209.

Learning objectives

On completion of this unit you should

1. be able to use the chain rule to find the derivative of a composite function;
2. be able to write down the derivative of any exponential function $f(x) = a^x$.

Remark: You do not have to prove the chain rule.

Assigned Problems

Stewart, Exercises 3.4 p 206: 3, all odd numbers from 7 to 21 and 39 to 51, 53, 58, 59, 65, 68, 69, 85, 90 a,b.

Unit 3.5 Implicit differentiation

Source: Stewart, Section 3.5 pp 209 - 216, 222 - 223.

Learning objectives

On completion of this unit you should

1. know when a function is defined implicitly;
2. be able to differentiate a function given implicitly by using implicit differentiation;
3. be able to use the technique and formulas in problems.

Remarks

1. You must be able to prove the formulas for the derivatives of the functions $y = \arcsin x$ and $y = \arctan x$, see pages 222-223.
2. You do not have to know the derivatives of the other inverse trigonometric functions (p 223).

Assigned Problems

Stewart, Exercises 3.5 p 214: All odd numbers from 5 to 21, 27, 31, 43, 47, 61, 64 and Exercises 3.6 p. 224: 63, 65, 69, 75.

Unit 3.6 Derivatives of logarithmic functions and inverse trigonometric functions

Source: Stewart, Section 3.6 pp 217 - 225.

Learning objectives

On completion of this unit you should

1. be able to write down and use the formula for the derivative of the logarithmic function $y = \log_a x$;
2. be able to write down and use the formula for the derivative of the logarithmic functions $y = \ln x$ and $y = \ln |x|$;
3. know when and how to use logarithmic differentiation; item be able to write down the formulas for the derivatives of the two inverse trigonometric functions $y = \arcsin x$ and $y = \arctan x$;
4. be able to use the formula and technique in problems.

Remarks

1. You must be able to prove the formulas for the derivatives of the logarithmic functions (pp 217-218).

2. The four cases given on p 221 are very important.
3. The following result is worth remembering:

$$e = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

4. You must be able to prove the formulas for the derivatives of the functions $y = \arcsin x$ and $y = \arctan x$.
5. You do not have to know the derivatives of the other inverse trigonometric functions (p 223).

Assigned Problems

Exercises 3.6 p 224: All odd numbers from 7 to 25, 32, 35, 39, 45, 51, 57.

Unit 3.7 Hyperbolic functions

Source: Stewart, Section 3.11 pp 261 - 268.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definitions of the hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$;
2. be able to sketch the graphs of the hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$;
3. be able to write down and use the derivatives of the functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$;
4. be able to use the formulas and graphs in problems.

Remarks

1. You should be able to prove the derivatives of the three hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$.
2. You do not have to memorize the identities on p 261. The identities can be proven by using the definitions of the two hyperbolic functions $\sinh$ and $\cosh$.
3. Omit the section on inverse hyperbolic functions.

Assigned Problems

Stewart, Exercises 3.11 p 266: Numbers 1, 2, 3, 27, 36, 37, 39.